# Harry Nguyen

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github.com/HarryNguyen2662 | timoto.ai
CS @ Georgia State (May 2027) · F-1 · OPT eligible

*New Grad Software Engineer — distributed systems, high-performance backend, and cloud infrastructure. Shipped C++ to Chrome stable (3B+ users); sole engineer on production AWS platform at sub-50ms p99.*

## Education

**Georgia State University** — Expected May 2027
Bachelor of Science in Computer Science · **GPA: 3.75/4.0** — Atlanta, GA

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures & Algorithms
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

## Work Experience

**Google** | Software Engineer Intern — Chrome Browser — May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec** — selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable
- Profiled settings navigation as p99 bottleneck at 1,200ms; implemented **lock-free concurrent trie search** eliminating mutex contention — **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern across **25K+ lines** of Chromium at **95% test coverage** — design documents adopted into production branch by senior Chrome engineers

**Develop for Good** | Software Engineer Intern — May 2024 – Aug 2024

- Designed stateless BaaS on **AWS** — JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned **PostgreSQL** indexing achieving **sub-100ms** for 10,000+ records

**CoderPush** | Software Engineer Intern — Sep 2023 – Dec 2023

- Diagnosed **DynamoDB** hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy — **30% read throughput improvement** under sustained concurrency
- Designed idempotent payment APIs with **Redis** distributed caching (**85% cache hit rate**) and exponential backoff — enforced exactly-once semantics under network partitions

## Personal Projects

**TiMoto AI — ML Serving Platform** | *vLLM, gRPC, Django, PostgreSQL, AWS, Terraform* — timoto.ai

- **Sole engineer** — designed, built, and operate end-to-end platform: vLLM inference, gRPC inter-service layer, Django REST backend, PostgreSQL, and multi-AZ AWS infrastructure
- Resolved **production deadlock under concurrent gRPC calls** — traced circular resource acquisition; redesigned call sequencing — **100% success rate** at **sub-50ms p99**
- Architected multi-AZ ECS Fargate with Terraform IaC and **circuit breaker pattern** — **99.9% uptime**, **44% cost reduction**; auto-rollback on health check failure

**Pulumi — Open Source Contributor** | *Go, TypeScript, IaC* — 24.4K★ github.com/pulumi/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization — studied linearizability guarantees under concurrent operations and partial failures

## Technical Skills

**Distributed Systems**: gRPC, Protocol Buffers, circuit breakers, exactly-once semantics, linearizability, Raft/Paxos, lock-free concurrency, event-driven architecture, fault tolerance
**Performance Engineering**: p99 latency profiling, mutex elimination, hot-partition diagnosis, sub-50ms SLOs, throughput optimization
**Languages**: C++, Python, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust
**Cloud & Infrastructure**: AWS (ECS Fargate, EC2, S3, RDS, DynamoDB), Docker, Kubernetes, Terraform, CI/CD, GitHub Actions
**Frameworks & Databases**: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB